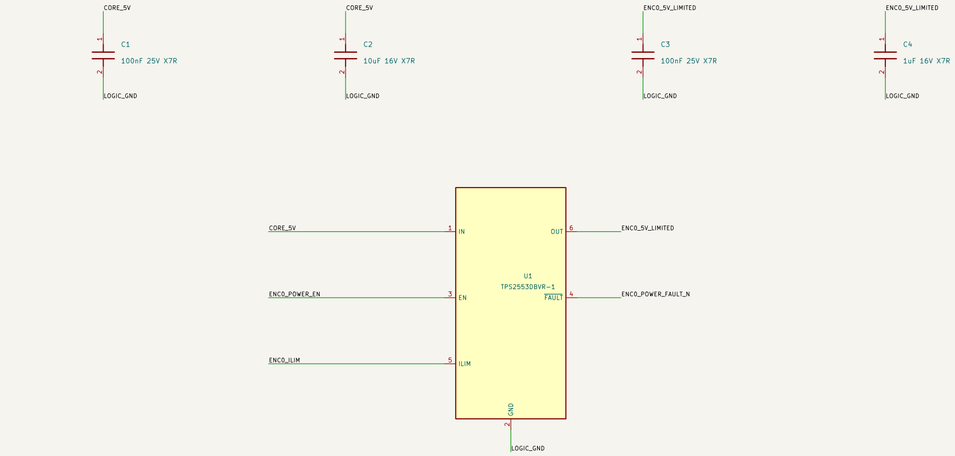
TWO CURRENT-LIMITED 5V ENCODER SUPPLIES — ENGINEERING BLOCK r0.1

TPS2553DBVR-1: DBV/SOT-23-6, not DRV. Exact receiver net names retained. No main supply/input protection, controller or board layout.

ENCODER 0: INDEPENDENT LATCH-OFF CURRENT LIMIT

CORE_5V required 4.75-5.25V; this switch does not regulate voltage.
200mA encoder load target; output at load depends on upstream and cable drop.



Place C1 at U1 IN/GND and R1 close to ILIM.
10k EN pulldown = off for a high-Z controller; no hardware restart latch drawn.
FAULT is open-drain, pulled only to IO_3V3; not an output-voltage/power-good measurement.



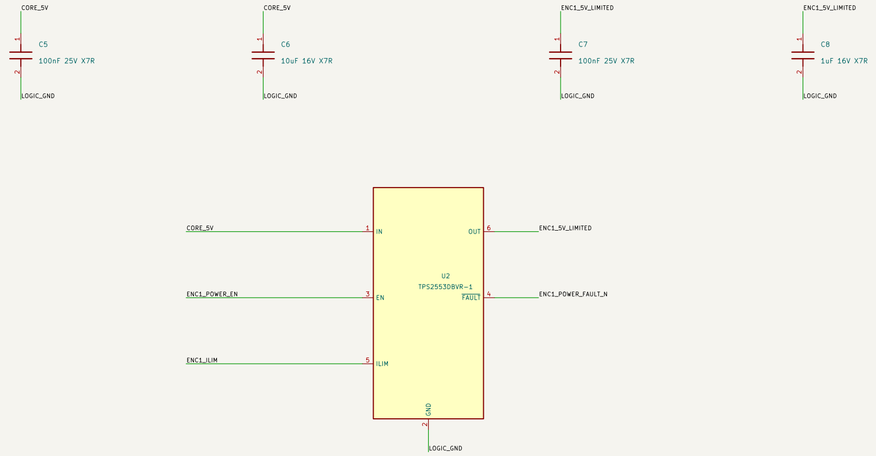
110k RILIM including 1%+150K/100ppm allowance: 207.445-283.605mA.
Two simultaneous pre-latch limits: 0.567210A plus switch supply current.
Ledger 0.4A is normal encoder load, not the fault budget. No full power budget approval.



Not a universal reverse-current barrier: reverse current can flow before protection responds.
No guarantee for 24V/reversed wiring, surge or system ESD. Board/cable faults need qualification.
100k output bleed is passive; no active discharge or guaranteed de-energization time.

ENCODER 1: INDEPENDENT LATCH-OFF CURRENT LIMIT

CORE_5V required 4.75-5.25V; this switch does not regulate voltage.
200mA encoder load target; output at load depends on upstream and cable drop.



Place C5 at U2 IN/GND and R5 close to ILIM.
10k EN pulldown = off for a high-Z controller; no hardware restart latch drawn.
FAULT is open-drain, pulled only to IO_3V3; not an output-voltage/power-good measurement.



Overcurrent/reverse events latch off after deglitch; EN toggle or power cycle clears latch.
THERMAL SHUTDOWN CAN AUTO-RETRY, including the -1 part. External fault-inhibit policy is missing.
A high EN retained across power return can energize the port. No automatic-restart prevention is claimed.



EN and FAULT interfaces target 0-3.6V, shared ground; partial-power/backfeed tests remain open.
Do not infer healthy power from an unasserted FAULT, especially with the switch/host unpowered.
Capacitor MPN/effective capacitance and startup/load tests are pending. Unreleased engineering source.

Engineering block; fault interface/physical qualification incomplete

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Title: Two protected encoder supply branches

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